

HARSH RAJ

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EDUCATION

Northeastern University, Boston, MA	May 2027 (expected)
Master of Science in Computer Science	GPA: 3.6/4.0
Delhi Technological University, Delhi, India	August 2023
Bachelor of Technology in Engineering Physics	GPA: 3.4/4.0

TECHNICAL SKILLS

Programming Languages:	Python, SQL, JavaScript, C++, Java
Cloud and DevOps:	AWS, GCP, Docker, Kubernetes, Git, CI/CD
Databases:	PostgreSQL, MySQL, MongoDB, Redis, Vector Databases
Tools and Libraries:	PyTorch, Jax, Hugging Face, Ray, LangChain, Pandas, NumPy, CUDA, Triton

PROFESSIONAL EXPERIENCE

Research Assistant, Forschungszentrum Jülich July 2025 – Present

- Contribute to EU Foundational Model projects ([OpenEuroLLM](#), [ELLIOT](#)) on JUPITER supercomputer
- Lead ablation studies on dataset mixtures and scaling laws on LLM training data at the petabyte scale
- Optimized multi-node data processing pipelines using Ray, achieving **5x** speedup in deduplication, filtering and map reduction operations

Co-Founder, Ontocord AI (Open Source) Jan 2024 – Jan 2025

- Co-founded open-source initiative focused on creating high-quality permissive datasets for LLMs
- Developed data processing pipelines with strategies for mixing, formatting, upsampling & interleaving
- Published [MixtureVitae-300BT](#) achieving 10% performance improvement over existing permissive data sets

Applied Scientist, Vijil AI Feb 2024 – Nov 2024

- Contributed to the [vijil-evaluation](#) platform for red-teaming of enterprise AI agent workflows. Integrated more than **10** benchmark datasets for vulnerability assessment
- Developed the vijil-fuzzer for adversarial text mutation. It achieved a 85% attack rate against GPT-4o

Machine Learning Engineer, Yield Inc Sept 2022 – Jan 2024

- Built synthetic data generation and finetuning infrastructure for LLM agents - [synchaeu](#) and [mandrill](#)
- Contributed to [Cacti](#), a Web3 transaction chatbot that simplifies blockchain interactions through natural language

SELECTED PUBLICATIONS

- Improving Consistency in Large Language Models through Chain of Guidance **1st Author** TMLR 2025
- Defences against Reverse Preference Attacks **3rd Author** AAI 2025
- Measuring Reliability of Large Language Models through Semantic Consistency **1st Author** NeurIPS 2022
Best Paper Award

PROJECTS

- terminal-bench** (600+ GitHub ★) Language agent sandbox evaluation framework (**Co-author & Maintainer**)
- inspect-ai** (1300+ GitHub ★) Industry-standard LLM evaluation suite (**Contributor**)

AWARDS AND ACHIEVEMENTS

- [SWISS AI](#) Grant Recipient – Awarded 200k GPU hours and CHF 61,000 in funding 2025
- [Rank 7](#) – Codeforces Raif ML Round 1 Competition 2021